

Chapter

Trigonometric Identities

1. Main 2023

$(1 + \sin A)(1 - \sin A)$ is equal to:

- (a) $\operatorname{cosec}^2 A$ (b) $\sin^2 A$
(c) $\sec^2 A$ (d) $\cos^2 A$

2. Comp 2023

Prove that : $\frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$.

3. Comp 2023

Prove that

$$(1 + \tan A - \sec A) \times (1 + \tan A + \sec A) = 2 \tan A$$

4. Main 2023

Prove the following identity

$$(\sin^2 \theta - 1)(\tan^2 \theta + 1) + 1 = 0$$

5. Main 2022 Sem II

Prove that : $1 + \frac{\tan^2 \theta}{1 + \sec \theta} = \sec \theta$

6. Main 2022 Sem II

Prove that : $\frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} = 2 \sec^2 \theta$

Solution

1.

$$(1 + \sin A)(1 - \sin A) = 1 - \sin^2 A$$

Using the identity $1 - \sin^2 A = \cos^2 A$,

$$(1 + \sin A)(1 - \sin A) = \cos^2 A$$

Correct answer: (d) $\cos^2 A$

2.

$$\frac{\sin A - 2 \sin^3 A}{2 \cos^3 A - \cos A} = \frac{\sin A(1 - 2 \sin^2 A)}{\cos A(2 \cos^2 A - 1)}$$

Since $1 - 2 \sin^2 A = 2 \cos^2 A - 1$,

$$= \frac{\sin A}{\cos A} = \tan A$$

Hence proved

3.

Yes, that's exactly what it means.

$$(1 + \tan A - \sec A)(1 + \tan A + \sec A) = \boxed{\text{RHS}}$$

After simplifying the LHS, we get:

$$(1 + \tan A)^2 - \sec^2 A = 1 + 2 \tan A + \tan^2 A - (1 + \tan^2 A) = 2 \tan A$$

So the correct answer is $\text{RHS} = 2 \tan A$ ✓

4.

$$(\sin^2 \theta - 1)(\tan^2 \theta + 1) + 1$$

Use identities:

$$\begin{aligned}\sin^2 \theta - 1 &= -\cos^2 \theta, & \tan^2 \theta + 1 &= \sec^2 \theta \\ &= (-\cos^2 \theta)(\sec^2 \theta) + 1\end{aligned}$$

Since $\cos^2 \theta \cdot \sec^2 \theta = 1$,

$$= -1 + 1 = 0$$

Hence proved

5.

$$1 + \frac{\tan^2 \theta}{1 + \sec \theta}$$

Use $\tan^2 \theta = \sec^2 \theta - 1$:

$$= 1 + \frac{\sec^2 \theta - 1}{1 + \sec \theta}$$

Factor numerator:

$$= 1 + \frac{(\sec \theta - 1)(\sec \theta + 1)}{1 + \sec \theta}$$

Cancel $(1 + \sec \theta)$:

$$\begin{aligned}&= 1 + (\sec \theta - 1) \\ &= \sec \theta\end{aligned}$$

Hence proved

6.

$$\frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta}$$

Take LCM:

$$\begin{aligned} &= \frac{(1 - \sin \theta) + (1 + \sin \theta)}{1 - \sin^2 \theta} \\ &= \frac{2}{\cos^2 \theta} \\ &= 2 \sec^2 \theta \end{aligned}$$

Hence proved
